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DETAIL SPECIFICATION

VALVE, NONFERROUS, FOR USE WITH HALOGENATED REFRIGERANTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonferrous manually operated valves for use with halogenated refrigerants.

1.2 Classification. Valves are of the following types and classes, as specified (see 6.2):

1.2.1 Type I – Diaphragm, nonback seating, packless

a. Class 1 – Shut-off service, angle, brazed
b. Class 2 – Shut-off service, straight, brazed
c. Class 3 – Shut-off service, angle, brazed to flare
d. Class 4 – Shut-off service, straight, flare
e. Class 5 – Shut-off service, straight, three-way flare
f. Class 6 – Throttling service, angle, brazed
g. Class 7 – Throttling service, straight, brazed
h. Class 8 – Charging, drain or purge, service, angle, brazed to flare
i. Class 9 – Charging, drain or purge, service, straight, brazed to flare

1.2.2 Type II – Packed, back seating

a. Class 1 – Shut-off service, angle, brazed
b. Class 2 – Shut-off service, straight, brazed

1.2.3 Type III – Diaphragm, back seating, packless

a. Class 1 – Shut-off service, angle, brazed
b. Class 2 – Shut-off service, straight, brazed
c. Class 3 – Shut-off service, angle, brazed to flare
d. Class 4 – Shut-off service, straight, flare
e. Class 5 – Shut-off service, straight, three-way flare
f. Class 6 – Throttling service, angle, brazed
g. Class 7 – Throttling service, straight, brazed
h. Class 8 – Charging, drain or purge, service, angle, brazed to flare
i. Class 9 – Charging, drain or purge, service, straight, brazed to flare

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.4 Type IV – Ball, two-port, inline body construction, shut-off service

a. Class 1 – Shut-off service, straight, brazed

b. Class 2 – Shut-off service, straight, welded

c. Class 3 – Charging, drain or purge, service, angle, brazed to flare

d. Class 4 – Charging, drain or purge, service, angle, weld to flare

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS B2.2/B2.2M	-	Specification for Brazing Procedure and Performance Qualification

(Copies of this document are available online at www.aws.org.)

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ASME

ASME B16.11 - Forged Fittings, Socket-Welding and Threaded

ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings

ASME B16.22 - Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings

ASME B16.25 - Buttwelding Ends

ASME B16.34 - Valves Flanged, Threaded and Welding End

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A276 - Standard Specification for Stainless Steel Bars and Shapes

ASTM A351/A351M - Castings, Austenitic, for Pressure-Containing Parts

ASTM B16/B16M - Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines

ASTM B36/B36M - Standard Specification for Brass Plate, Sheet, Strip and Rolled Bar

ASTM B62 - Standard Specification for Composition Bronze or Ounce Metal Castings

ASTM B98/B98M - Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes

ASTM B121/B121M - Standard Specification for Leaded Brass Plate, Sheet, Strip and Rolled Bar

ASTM B124/B124M - Standard Specification for Copper and Copper-Alloy Forging Rod, Bar, and Shapes

ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

ASTM F104 - Standard Classification System for Nonmetallic Gasket Materials

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS) OF THE VALVE AND FITTING INDUSTRY

MSS SP-110 - Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

(Copies of this document are available online at www.mss-hq.org.)

SAE INTERNATIONAL

SAE-J513 - Refrigeration Tube Fittings – General Specifications

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. All material used in the construction of the valves shall be as specified herein. Other materials not specified may be selected if the procuring activity has been satisfied by testing that the proposed substitutes fully meet the design and service requirements of this specification, including halogenated refrigerant compatibility, brazing or weldability, and corrosion resistance without requiring painting, coating, or plating. Materials for contacting parts shall be selected to minimize electrolytic corrosion, galling, seizing, and excessive wear. All stainless steels are considered of a similar galvanic series. Nickel-copper, copper-nickel, nickel-aluminum-bronze, copper, bronze, and brass are considered of a similar galvanic series.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Bodies, bonnets, caps, and end-connections.

3.2.2.1 Types I and III. For Types I and III valves, bodies, bonnets, or caps up to and including ⅝ inch shall be constructed of forged brass in accordance with ASTM B124/B124M, Alloy 377. Valves above ⅝ inch may be forged or cast brass. Cast valves shall be of brass in accordance with ASTM B584, Alloy 4A. End connection or tail-piece material shall be the same material as the body of the valve or shall be of a similar galvanic series.

3.2.2.2 Type II. For Type II valves, bodies, bonnets, or caps shall be constructed of forged brass or cast bronze in accordance with ASTM B124/B124M, Alloy 2, or ASTM B62, as applicable. Bonnets shall be of the flange type or such other types as may be determined suitable for military use. End connection or tail-piece material shall be the same material as the body of the valve or shall be of a similar galvanic series.

3.2.2.3 Type IV. For Type IV valves, the pressure containing boundary and all internal parts in contact with the halogenated refrigerant shall be of the following corrosion-resistant materials: stainless steel (ASTM A182/A182M, ASTM A276, and ASTM A351/A351M – S30200, S30400, or S31600 of the unified numbering system), nickel-copper (QQ-N-281), copper-nickel (MIL-C-15726, ASME B124), nickel-aluminum-bronze (QQ-N-286 and not less than 4 percent nickel content), copper (ASTM B98/B98M, ASTM B124/B124M, and ASTM B584), bronze (ASTM B62 and ASTM B584), or brass (ASTM B121/B121M and ASTM B584). End connection or tail-piece material shall be the same material as the body of the valve or shall be of a similar galvanic series.

3.2.3 Handwheels. Types I, III, and IV valves shall be provided with either a fixed or removable handwheel as specified (see 6.2) and constructed of corrosion-resistant metal, a die cast alloy with a suitable protective finish, or malleable iron with a suitable protective coating. Types I and III valves may use removable color-coded (see 6.2) thermal plastic handwheels with a continuous service temperature of not less than 200 °F and a melting point of not less than 450 °F. Loss of a thermal plastic handwheel shall not prevent operation of the valve with crescent or similar flat-sided wrench. Valve handwheels shall be sized to be operated single handed with a maximum operating torque of 10 foot-pounds.

3.2.4 Protection against corrosion. In order to prevent deterioration due to corrosion; bolts, nuts, studs, pins, screws, and such other fastenings or fittings as may be required shall be of corrosion-resisting steel, copper, brass, bronze, copper-nickel, copper-silicon, nickel-copper, or nickel-aluminum-bronze alloy in accordance with ASTM A276; ASTM B124/B124M; ASTM B16/B16M; ASTM B36/B36M; MIL-C-15726; ASTM B98/B98M, Alloy B; QQ-N-281; and QQ-N-286, respectively.

3.2.5 Material for internal parts and valve stems. All internal parts of valves such as seats, disks, stems, and springs that come in contact with the refrigerant shall be of a corrosion-resistant material. The treatment of ferrous material by plating is not considered satisfactory as a corrosion-resistant material for internal parts. Valve stems shall be of a material and hardness to prevent galling.

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3.2.6 Nonmetallic materials. All internal nonmetallic materials shall be compatible with procuring activity specified refrigerant and associated lubricating oil (see 6.2).

3.2.7 Prohibited materials.

3.2.7.1 Asbestos. Asbestos and material containing asbestos are prohibited.

3.2.7.2 Cadmium. The use of cadmium plating is prohibited unless approved by the procuring activity.

3.2.7.3 Mercury. Mercury and mercury containing instruments (except fluorescent or necessary vapor lighting) shall not be used in the manufacture or testing of any item listed by this specification. Mercury or mercury containing compounds shall not be intentionally added or come in direct contact with hardware or supplies furnished under this specification.. Hardware is mercury-free if it does not contain functional mercury and is not contaminated by mercury or mercury compounds. Functional mercury is mercury or mercury compound(s) contained in equipment that is required for the equipment to operate properly.

3.3 Sizes. Valves shall be of the sizes shown in tables I through V, as specified (see 6.2).

TABLE I. Types I and III valve sizes.

Vertical center line of valve to face of side connection						
Valve size	Class 1		Classes 3 and 8		Class 6	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
¼	1 5/16	1 7/16	1 9/32	1 5/16	1 5/16	1 7/16
⅜	1 5/16	1 7/16	1 5/16	1 7/16	1 5/16	1 7/16
½	1½	1 15/16	1 9/16	1 7/8	1 5/16	1 15/16
⅝	1½	1 15/16	1 11/16	1 15/16	1 5/8	1 15/16
⅞	2⅛	2¼	---	---	1 7/8	1 15/16
1⅛	2½	2 9/16	---	---	2⅛	2 3/16

NOTES:
1. All dimensions are in inches.
2. Tolerances are ±1/32.

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TABLE II. Type II valve sizes.

Valve size	Class 1	
	Minimum	Maximum
7/8	1 13/16	2
1 1/8	2	2 11/32
1 3/8	2 3/8	2 23/32
1 5/8	2 17/32	3
2 1/8	3 3/16	3 5/8
2 5/8	3 19/32	4 5/16
3 1/8	4 1/16	4 5/8

NOTES:
1. All dimensions are in inches.
2. Tolerances are ±1/32.

TABLE III. Types I and III valve end-to-end dimensions.

Valve size	End-to-end dimension								End-to-end dimension (run)		Center line of run to face of branch connection	
	Class 2		Class 4		Class 7		Class 9		Class 5		Class 5	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
1/4	2 5/8	2 25/32	2 5/8	2 25/32	2 5/8	2 25/32	2 5/8	2 25/32	2 3/4	3 3/8	1 3/8	1 9/16
3/8	2 5/8	2 25/32	2 5/8	2 25/32	2 5/8	2 25/32	2 5/8	2 25/32	2 3/4	3 7/16	1 3/8	1 11/16
1/2	3 1/8	3 11/16	3 1/8	3 1/2	2 5/8	3 11/16	3 1/4	3 1/2	3	3 23/32	1 3/4	1 7/8
5/8	3 1/2	4	3 5/8	4	3 3/8	4	3 5/8	3 7/8	3 3/8	3 7/8	1 5/16	2 3/16
7/8	4 3/8	4 13/16	---	---	4 7/16	4 1/2	---	---	---	---	---	---
1 1/8	5 3/16	5 15/16	---	---	4 3/8	4 15/16	---	---	---	---	---	---

NOTES:
1. All dimensions are in inches.
2. Tolerances are ±1/32.

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TABLE IV. Type II valve end-to-end dimensions.

End-to-end dimension		
Valve size	**Class 2**	
	Minimum	**Maximum**
7/8	4¼	4¾
1⅛	4⅞	5 3/16
1⅜	5¼	6 7/16
1⅝	6½	6¾
2⅛	8	8½
2⅝	9⅝	11
3⅛	11	12
3⅝	12¾	14
4⅛	14	15

NOTES:
1. All dimensions are in inches.
2. Tolerances are ±1/32.

TABLE V. Type IV, Class 1 valve dimensions.

Valve size NPS (nominal pipe size)	**End-to-end dimension for butt-welded, socket-welded, and solder joint end connection**	
	Minimum	**Maximum**
¼	2	2½
⅜	2	2½
½	2½	3
¾	2½	3
1	3¼	4¼
1¼	4¼	4⅝
1½	4¼	4⅝
2	4⅞	5⅞

NOTES:
1. All dimensions are in inches.
2. Tolerances are ±1/32.

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3.4 Design. Valves furnished under this specification shall incorporate the applicable design features of ASME B16.34 and MSS SP-110 and shall be based on service proven designs. Valves shall be assembled by mechanically threaded bolts, brazing, or welds. The use of soft-solder joints is prohibited. Valves shall be assembled by mechanically threaded joints, bolts, brazing, or welds. The use of soft-solder joints is prohibited. Where brazed joints are used, the brazing shall be in accordance with AWS B2.2/B2.2M, or equal. Brazing shall use cadmium-free silver solder unless the material combination requires cadmium silver solder as specified by AWS B2.2/B2.2M. Welding shall be approved by the procuring activity (see 6.2).

3.4.1 Valve stems. The valve shall close by a clockwise rotation of the handwheel and/or handle when viewed from directly above the handwheel and/or handle. Types I, II, and III valves shall be provided with a rising stem to indicate open or closed position. Stem disks shall not rotate on the seat during closing or opening. The operating screw (upper stem) interface with the diaphragm during closing or opening shall meet the cyclic endurance requirements of 4.3.2. The operating screw (upper stem) and threaded valve bonnet shall be of materials having a sufficient difference in hardness to minimize thread galling. The operating screw (upper stem) shall back seat on the valve bonnet. The lower stem on Type I valves may or may not back seat when the valve is completely open. The lower stem on Type II valves shall back seat against the bonnet when the valve is completely open. The lower stem on Type III valves shall back seat against the body or body bushing when the valve is completely open. The lower stem assembly shall be screwed into the valve body with a shouldered seal. All valve seat disks shall be securely retained and fastened to the lower stem by crimping or other mechanical means such as locking screws or nuts. Where seat disks are retained by crimping only, a suitable vent shall be provided between the seat disk and lower stem to discharge any entrapped refrigerant to prevent possible dislodgment of the seat disk. Provision shall be made to permit opening of the valve under pressure without sticking of the stem and seat.

3.4.2 Orifice. Valves, except those used for throttling or expansion purposes, shall have an unrestricted flow area at least equivalent to the area of a circle having a diameter equal to the nominal pipe size or tube inside diameter. Type IV valve port dimensions shall be as specified in MSS SP-110, or as specified (see 6.2).

3.4.3 Connections. Valves shall have end connections selected from those listed in table VI and as specified in 6.2.

TABLE VI. End connections for valves.

Type	**Applicable document for dimensional details**
Butt-welded	ASME B16.25
Socket-welded	ASME B16.11
Solder Joint – casting	ASME B16.18
Solder Joint – forging	ASME B16.22
Refrigeration flare	SAE-J513
Other, as specified	As specified

3.4.4 Threads. Threads shall conform to the requirements of FED-STD-H28. Tapered pipe threads shall not be used. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. The design shall be such that standard wrenches can be used on all external bolting. Lock wire shall not be used. Any exposed threads shall be protected by plastic caps for shipping.

3.4.5 Fasteners. Fastener strength factors and length of engagement shall conform to the requirements of FED-STD-H28. Fastener material shall be corrosion-resistant as approved by the procuring activity. Where studs or bolts with nuts are used, the length of the stud or nut shall be sufficient so that after assembly, a minimum of one thread and a maximum of ten threads shall protrude beyond the crown of the nut.

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3.4.6 Seal caps. Valves provided with SAE-J513 flare end connections shall be provided with caps to seal the charging and purging end-connection. Type II valves shall be provided with a cap to seal around the packing and stem. Type III valves shall be provided with a cap to seal the stem if operational testing (see 4.3.2) shows stem leakage. All seal caps shall be designed to meet the pressure rating of the valve. If the seal cap uses a gasket for a pressure boundary, the gasket shall be a retained type to prevent the loss of the gasket, and the gasket shall be replaceable. The seal cap shall be:

a. Configured to be readily tightened or loosened by hand.

b. Fabricated of metallic or durable non-metallic materials.

c. Capable of withstanding normal abuse and wear.

d. Equipped with a stay chain.

3.4.7 Position indicator. Type IV valves shall incorporate a means of indicating the ball position in either a 90 degree or 180 degree operation. The means of indication (travel stop) shall be governed by the following: the ball position shall be indicated by the words “OPEN” and “CLOSED” on the valve bonnet (where applicable). Where a handle is used, the handle shall be parallel to the connecting pipe axis to indicate when the ball is fully open. Where a handwheel is used, a groove (parallel with the ball port axis) or other means of position indication shall be attached directly to the valve stem to indicate when the ball is fully open or closed.

3.4.8 Thrust washer. When thrust collars are incorporated into a valve, a thrust washer shall be employed in conjunction with the thrust collar to facilitate valve operation. This washer shall be self-lubricating and renewable.

3.4.9 Maintenance. Valves shall be of such a design that the internal working parts may be removed for inspection, repair, or replacement without the necessity of applying heat to the valve or breaking line connections.

3.4.10 Valve seat. Bodies of Type II and IV valves shall be so fitted that the valve seat may be easily refaced or replaced. For Type IV valves, soft-seating (non-metallic) inserts shall be field replaceable. Any soft-seating insert shall be protected from excessive loading and extrusion or from any other effect jeopardizing their useful life. Soft-seating inserts shall be of the simplest practical configuration to facilitate emergency replacement manufacture where necessary. Seat inserts shall not be physically reversible unless they are also functionally reversible to preclude incorrect assembly. The center section of Type IV valves shall be able to swing out or be removed to allow for replacement of all component parts without disturbing pipe alignment.

3.4.11 Mounting. Mounting flanges or lugs shall be provided on all straight Type I, II, and III valves.

3.4.12 Type II valve, packed, back seating.

a. Valves shall be globe valves provided with wing caps.

b. Valves shall be of the back seating type to permit repacking under pressure. The valve stem stuffing box and gland shall be designed so that the packing may be replaced while the valve is under pressure. Packing shall be non-metallic molded rings or non-metallic continuous strip and shall be suitable for use with halogenated refrigerants and associated lubricating oils. Material deemed equally effective may be used, as approved by the procuring activity.

c. Retained type gaskets shall be used in all seal caps to prevent loss. Bonnet gaskets shall be made of compressed non-asbestos fiber material conforming to ASTM F104. Material deemed equally effective may be used, as approved by the procuring activity.

3.5 Marking. Valves shall have the following information distinctly cast or stamped on the side of the body or on a plate secured to the top of the handwheel. Type I valves, Classes 6 and 7, shall in addition show the letters “EXP”. Valve handwheels or plates attached thereto shall show a directional arrow with the word “OPEN”. Where a plate under the handwheel is used for the marking, it shall neither project outside the handwheel nor jut in such a manner as to be dangerous to handle. Valves shall have distinctly cast or stamped on one side of the body an arrow indicating direction of flow to show that the inlet pressure is under the valve seat (except for Type III valves that are designed to be non-directional).

a. Trademark or name of manufacturer.

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b. Manufacturer part number and size of valve. Manufacturer part number will clearly indicate any temperature and refrigerant compatibility restrictions.

c. Pressure rating.

3.6 Pressure and temperature rating. Type I, II, and III valves shall have pressure-temperature ratings that meet or exceed the requirements specified in table VII for refrigeration. Type IV valves shall, at a minimum, have pressure-temperature ratings that meet or exceed the requirements specified in table VII for air conditioning. The supplier may offer a Type IV valve that meets the requirements specified in table VII for refrigeration. The inlet(s) and outlet(s) pressure and temperature ratings of the valve shall be identical for any given valve. The pressure-temperature ratings of valves shall be determined by the material of the valve body, seats, stem seals, end connections, or any other type of construction that would be restrictive. Subject to the valve material, pressure-temperature ratings shall not exceed those specified in ASME B16.18, ASME B16.22, or ASME B16.34, as applicable, without approval of the procuring activity. Pressure ratings shall be verified as specified in 3.8.1 through 3.8.3.

TABLE VII. Pressure and temperature ratings.

Application	Temperature rating		Pressure rating	
	Minimum (°F)	Maximum (°F)	Minimum (mmHg)	Maximum (psig)
Air conditioning (including chilled water)	25	200	0.5 (500 microns)	300
Refrigeration (food storage)	-40	200	0.5 (500 microns)	300

NOTES:

1. Valve burst pressure rating shall be not less than four times the maximum working pressure marked on the valve.
2. Valve hydrostatic strength rating shall be 150 percent of the rated maximum working pressure or 450 psig, whichever is higher.
3. The valve labeled maximum working pressure shall be not less than 110 percent of the applicable refrigerant saturation pressure at 150 °F.

3.7 Environmental requirements.

3.7.1 High shock. Valves shall be designed to withstand the Grade A shock requirements in accordance with MIL-S-901. Valves shall be considered to have failed the shock tests in the event of any of the following:

a. Breakage of any parts, including mounting bolts.

b. Appreciable distortion or dislocation of any part or otherwise creating a hazard to personnel or other equipment. Included is binding of the handwheel that would exceed the operating torque specified in 3.2.3.

c. Significant effect on performance when tested as specified in 4.4.3 and 4.4.4.

3.7.2 Vibration. Valves shall be designed to withstand vibration and shall have successfully met the Type I vibration requirements of MIL-STD-167-1. Equipment shall be considered to have failed the vibration tests in the event of any of the following:

a. Breakage of any parts, including mounting bolts.

b. Appreciable distortion or dislocation of any part or otherwise creating a hazard to personnel or other equipment. Included is binding of the handwheel that would exceed the operating torque specified in 3.2.3.

c. Significant effect on performance when tested as specified in 4.4.3 and 4.4.4.

3.8 Performance requirements. All valves shall perform without refrigerant leakage defined in table VIII.

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TABLE VIII. Leakage criteria.

Test fluid (see Note)	Leak detection method	Leak acceptance criteria
Nitrogen	Submerge in clear liquid	No bubble forms and flows away in 4 minutes
Nitrogen-refrigerant trace gas	Electronic detector	½-ounce refrigerant per year
Nitrogen-helium trace gas	Electronic detector	½-ounce refrigerant per year equivalent

NOTE:
1. Trace gas shall contain refrigerant or helium not less than that required to raise pressure 10 psi.

3.8.1 Strength pressure. The valve shall exhibit no signs of leakage, porosity, or deformation when tested in accordance with 4.4.3.

3.8.2 Refrigerant leakage. Valves shall show no evidence of refrigerant leakage as defined by table VIII when tested in accordance with 4.4.4.

3.8.3 Vacuum hold. Valves shall be vacuum-tight with no change in pressure when tested in accordance with 4.4.5.

3.8.4 Reliability. The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirement of design. Where wear is unavoidable, the parts subjected to much wear shall be of the best material for the purpose in order to reduce these detrimental effects to a minimum. Valves subject to external condensation shall consider the detrimental effects of corrosion and select materials that will maximize reliability. Valves shall meet an open-close cycle reliability when tested as specified in 4.4.6. The valve operating torque shall not exceed that specified in 3.2.3. The valve shall exhibit no signs of failure or refrigerant leakage as defined by table VIII. The valve after completing the specified number of cycles shall be vacuum-tight with no change in pressure for a period of not less than 15 minutes.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be accomplished on one valve of each type and class in accordance with the examination and tests specified in 4.3 and 4.4.

4.3 Conformance inspection. Conformance inspection shall be performed to verify that the valve meets specification requirements prior to acceptance and shall include the examinations specified in 4.3.2 and tests specified in 4.4.3 and 4.4.4.

4.3.1 Lot. All valves of the same type, class, and size that are presented at one time shall be considered a lot for purposes of conformance inspection.

4.3.2 Examination. Each sample valve selected in accordance with table VIII shall be visually examined for compliance with this specification as follows:

a. Materials (see 3.3).

b. Piping connection dimensions and tolerance (see 3.3).

c. Design of valve (see 3.4).

d. Marking (see 3.5).

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4.3.3 Sampling for examination and operating test. A random sample of valves shall be selected in accordance with table IX from each lot for examination specified in 4.3.2 and tests specified in 4.4.3 through 4.4.4. Any valve in the sample containing one or more defects shall not be offered for delivery.

TABLE IX. Sampling for examination and test.

Number of valves in lot	Number of valves in sample
15 and under	10
16 to 40	15
41 to 110	25
111 to 300	35
301 to 500	50
501 to 800	75
801 to 1,300	110
1301 and over	150

4.4 Tests.

4.4.1 High impact shock test. One complete valve shall be subjected to a shock test specified in MIL-S-901 for Grade A, Class I equipment for hull and deck mount with unrestricted orientation. The largest size valve of each type shall be tested as approved by the procuring activity.

4.4.2 Vibration test. One complete valve shall be subjected to the Type I vibration tested 4-33 Hz as specified in MIL-STD-167-1.

4.4.3 Strength pressure test. The valve shall be strength tested with water or test fluid listed in table VIII at a pressure of not less than 450 psig, or as specified (see 6.2), for not less than 5 minutes. The test shall be made initially with the test pressure applied under the closed valve to determine leakage of the seat and disk. If there is no evidence of leakage at the valve seat, the valve shall be open but not backseated and the pressure applied to determine the soundness and strength of the complete valve.

4.4.4 Refrigerant leak test. Each valve shall be pressurized with test fluid as specified in table VIII to not less than 300 psig, or as specified (see 6.2). The test pressure shall be applied above the valve disk with the valve closed to determine leakage of the seat and disk and in the bonnet and cap assemblies. Additionally, the entire valve shall be subjected to the test pressure to determine leakage and porosity.

4.4.5 Vacuum hold test. With the valve in the partially open position, the minimum pressure (vacuum) specified table VIII shall be drawn for a period of not less than 15 minutes at the valve inlet(s), with the outlets(s) blanked off.

4.4.6 Reliability test. The valve, pressurized with test fluid as specified in table VIII to not less than 300 psig, or as specified (see 6.2), shall be cycled and tested as follows:

a. Cycle 50 times and then seat valve with the seating force as specified by the manufacturer. One cycle shall consist of one complete opening and closing during a period not to exceed 15 seconds. Seating, unseating, and running torques and stem leakage shall be noted when cycling the valve.

b. After 50 cycles, the valve shall remain pressurized for 5 minutes in the seated condition, and then seat leakage shall be checked. Seat leakage testing shall be accomplished at not more than the maximum torque specified by the manufacturer.

c. Repeat a and b until the valve has undergone the number of cycles specified in table X. At the completion of the required cycles, the valve shall be checked for vacuum tightness as specified in 4.4.5.

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TABLE X. Valve operating cycles.

Valve type	Valve size	Number of cycles
I and III	<½ inch	2500
	½ inch to ⅝ inch	2000
	>⅝ inch	900
II and IV	All	2500

TABLE XI. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Examination	X	X	3.2, 3.3, 3.4, 3.5	4.3.2
High shock	X		3.7.1	4.4.1
Vibration	X		3.7.2	4.4.2
Strength pressure	X	X	3.6, 3.8.1	4.4.3
Refrigerant leakage	X	X	3.6, 3.8.2	4.4.4
Vacuum hold	X	X	3.6, 3.8.3	4.4.5
Reliability	X		3.8.4	4.4.6

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Naval Supply Command's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Valves detailed by this specification are intended for use by vapor compression chilled water plants and air conditioning and refrigeration equipment that are required to be shock and vibration qualified.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size of valves required (see 1.2 and 3.3).

c. When first article is required (see 3.1).

d. Whether a fixed or removable handwheel is required and if the handwheel should be color-coded (see 3.2.3).

e. Refrigerant type and associated lubricating oil (see 3.2.6).

f. Size (see 3.3).

g. Welding requirements (see 3.4).

h. Type IV valve port dimension, if other than as specified in MSS SP-110 (see 3.4).

i. Refrigerant(s) application and test pressure (see 4.4.3 and 4.4.4).

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j. Packaging requirements (see 5.1).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Testing site. Where a manufacturer is unable to conduct these tests at their own plant, they may arrange to have them conducted at a commercial laboratory or Government laboratory suitably equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable approved drawings should accompany the valves.

6.5 Subject term (key word) listing.

Ball valve

Diaphragm packless

Manual valve

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR4
Navy – SH
Air Force – 99
DLA – GS

Preparing activity:
Navy – SH
(Project 4130-2014-002)

Review activities:
Army – GL
Navy – YD
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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